INTRODUCTION TO BUSINESS INTELLIGENCE AND POWER BI

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FSDM

OUTLINE

- ❑ Data and Decision-Making
- ❑ What is Data Analysis?
- ❑ What is Business Intelligence (BI)?
- ❑ BI vs Data Analysis
- ❑ Tools Used in This Course
- ❑ What You Will Learn in This Course

DATA AND DECISION-MAKING

- Why Data Matters in Organizations
  - Organizations generate large amounts of data every day
  - Decisions should be based on evidence, not intuition
  - Data allows organizations to measure performance using KPIs
  - Reliable and structured information leads to better decisions

- From Data to Decision
  - Raw Data → Preprocessing (Cleaning) → Analysis (Context) → Information → Insights → Decisions

WHAT IS A DATA ANALYSIS?

- Data analysis is the process of transforming raw data into useful insights to support decision-making.

- It starts with a clear question and ends with conclusions that help solve a problem.

WHAT IS A DATA ANALYSIS?

- Main Stages of Data Analysis

1. Define the Question
2. Collect the Data
3. Clean and Prepare the Data
4. Analyze the Data
5. Interpret and Report Results

WHAT IS A DATA ANALYSIS?

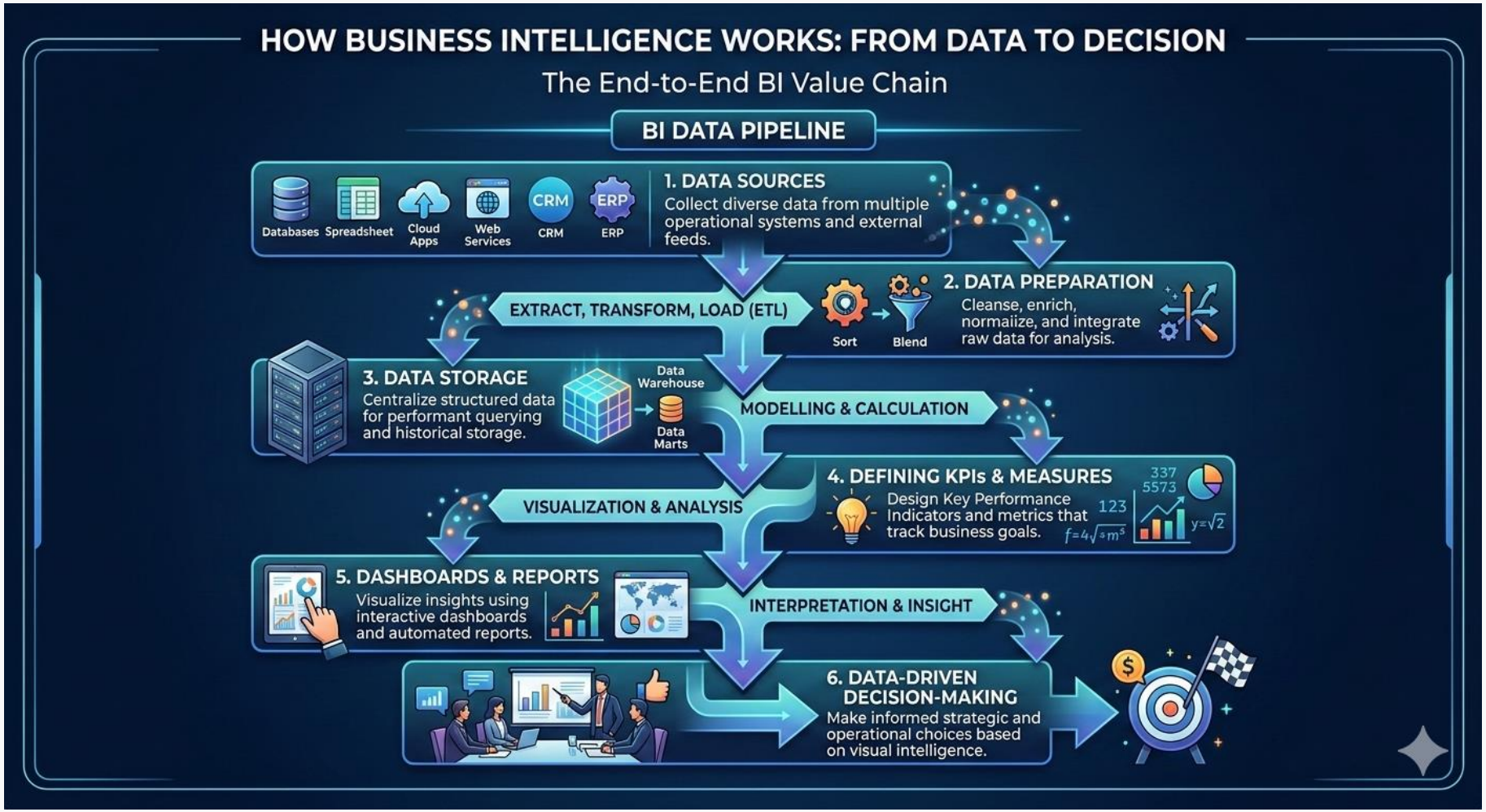
- Define the Question: Translate a business problem into measurable indicators.
- Collect the Data: Identify and gather relevant and reliable data sources.
- Clean and Prepare: Handle missing values, errors, and inconsistencies.
- Analyze: Apply statistical or analytical methods to extract patterns.
- Interpret & Report: Present results clearly to support decision-making.

WHAT IS BUSINESS INTELLIGENCE?

- Business Intelligence (BI) is the set of tools and processes used to transform organizational data into useful information for decision-making.

- BI helps organizations:
  - Monitor performance
  - Measure KPIs
  - Identify trends
  - Support operational and strategic decisions

WHAT IS BUSINESS INTELLIGENCE?



WHAT IS BUSINESS INTELLIGENCE?

- Main Components of a BI System
  - Data Sources (ERP, CRM, files, databases)
  - Data Preparation & Transformation
  - Storage (structured repository)
  - KPI & Metric Definition
  - Visualization (Dashboards)
  - Decision & Feedback

BUSINESS INTELLIGENCE VS DATA ANALYSIS

Aspect	Data Analysis	Business Intelligence
Goal	Explore data to discover patterns and insights	Monitor performance and support decisions
Focus	Deep investigation of datasets	Structured monitoring using KPIs
Users	Data analysts, data scientists	Managers, decision-makers
Time Perspective	Often exploratory and project-based	Continuous monitoring over time
Tools	Python, R, statistical tools	BI platforms (Power BI, Tableau)

TOOLS USED IN THIS COURSE

- This course uses three main tools:
  - Python with Pandas
  - Polars
  - Power BI Desktop
- These tools cover the main stages of the data analysis workflow:
  - Data preparation
  - Data analysis
  - Data visualization and reporting

TOOLS USED IN THIS COURSE

- Python (programming language) is widely used in data science.
- Pandas (software library) is a Python library designed for:
  - Data cleaning
  - Data transformation
  - Data exploration
  - Statistical analysis
- It was created by Wes McKinney in 2008 and is one of the most widely used tools in data analysis.

TOOLS USED IN THIS COURSE

- Polars (dataframe library) is a modern data processing library.
- Key characteristics:
  - Very fast data processing
  - Designed for large datasets
  - Parallel execution (multi-core CPUs)
- Polars is written in Rust (programming language) and was created by Ritchie Vink in 2020.

TOOLS USED IN THIS COURSE

Feature	Pandas	Polars
Language	Python	Rust (Python API available)
Performance	Good, but slower on large datasets	Very fast, optimized for large datasets
Execution Model	Eager execution	Eager + Lazy execution
Parallel Processing	Mostly single-threaded	Multi-threaded by default
Memory Layout	Row-oriented (NumPy-based)	Columnar (Apache Arrow format)
Memory Efficiency	Moderate	More memory efficient
Handling Big Data	Can struggle with very large datasets	Designed for large-scale data processing
Ease of Use	Very beginner-friendly	Slightly steeper learning curve
Ecosystem	Huge ecosystem, very mature	Growing ecosystem
Best For	General data analysis, education	High-performance analytics, data engineering

Pandas vs Polars Comparison

TOOLS USED IN THIS COURSE

- Microsoft Power BI Desktop is a Business Intelligence tool used to:
  - Connect to different data sources
  - Transform and model data
  - Create dashboards and reports
  - Share insights with decision-makers
- Power BI is widely used for business reporting and KPI monitoring.

TOOLS USED IN THIS COURSE

WHAT YOU WILL LEARN IN THIS COURSE

- By the end of this course, you will be able to build a complete data analysis and reporting workflow using Python and Power BI.

WHAT YOU WILL LEARN IN THIS COURSE

1. **Data Cleaning and Preparation with Python**

- Using Python (programming language) with pandas (software library) and Polars (dataframe library) to:
  - Handle missing values and inconsistent data
  - Transform and normalize datasets
  - Perform aggregations and data restructuring
  - Work with large datasets efficiently
  - Compare performance between Pandas and Polars on large datasets

WHAT YOU WILL LEARN IN THIS COURSE

2. Data Transformation with Power Query

- Using Power Query in Microsoft Power BI Desktop to:
  - Clean and transform datasets
  - Pivot, unpivot, transpose, and reshape tables
  - Fill Down / Fill Up
  - Conditional columns
  - Split and merge columns
  - Joins (Merge Queries) and Append
  - Execute Python scripts inside Power Query
  - Understand the basics of **M language** generated by Power Query

WHAT YOU WILL LEARN IN THIS COURSE

3. Data Integration from Multiple Sources

- Loading and integrating data from:
  - CSV and Excel files
  - MySQL databases
  - Online sources such as Wikipedia tables

WHAT YOU WILL LEARN IN THIS COURSE

4. Data Modeling and Query Modes in Power BI

- Understanding how Power BI works with data:
  - Import mode
  - DirectQuery mode
  - Live connections

WHAT YOU WILL LEARN IN THIS COURSE

5. Dashboard Design and Visualization

- Using Power BI Desktop to:
  - Create charts, tables, and interactive visuals
  - Apply filters and slicers
  - Design multi-page reports

WHAT YOU WILL LEARN IN THIS COURSE

6. Analytical Measures with DAX

- Using DAX (Data Analysis Expressions) to:
  - Create calculated measures
  - Define business metrics and KPIs
  - Implement time intelligence calculations

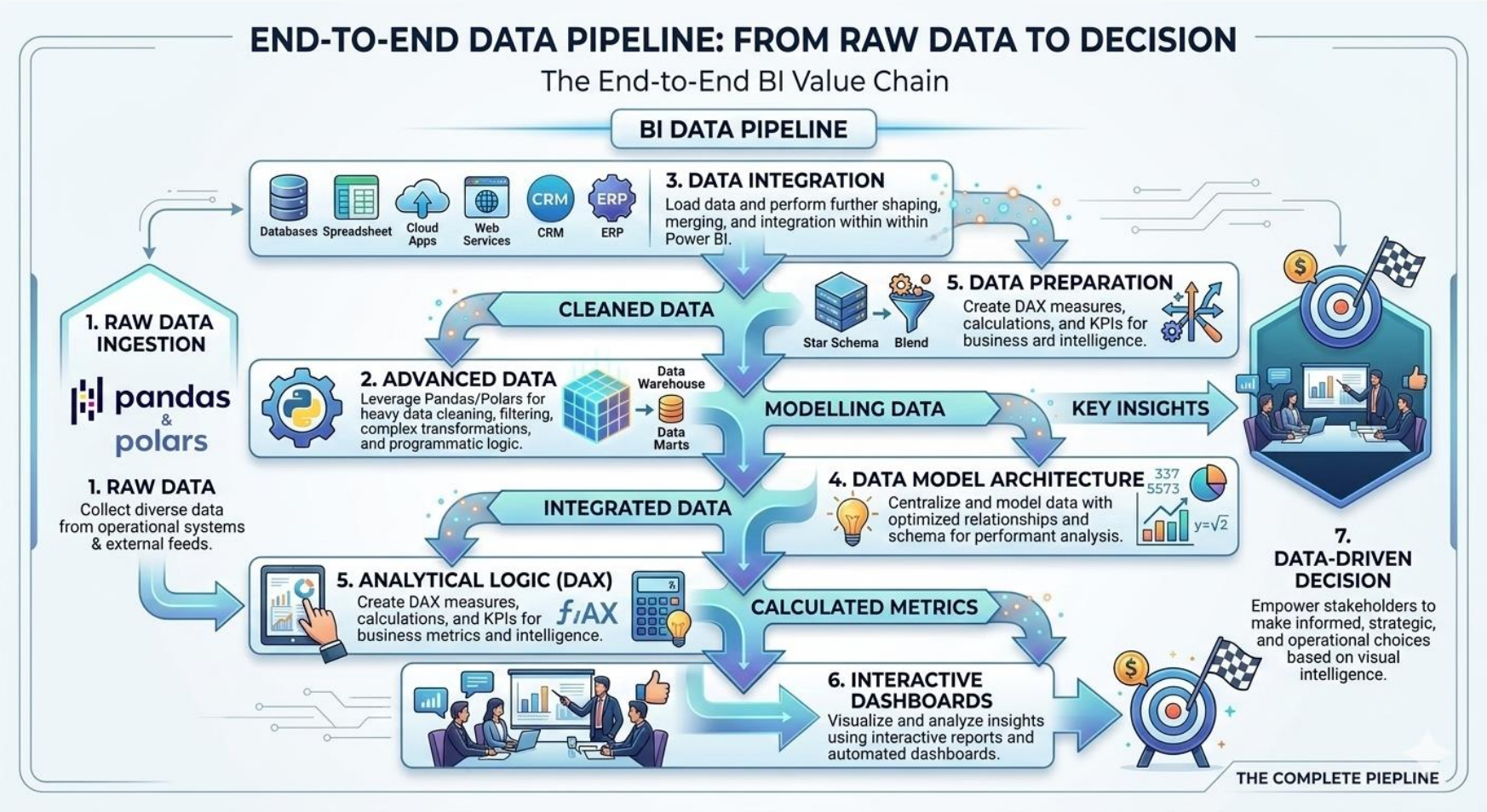
WHAT YOU WILL LEARN IN THIS COURSE

7. Interactive Reporting

- Building interactive dashboards using:
  - Bookmarks and buttons
  - Navigation between report pages
  - Dynamic visual interactions

WHAT YOU WILL LEARN IN THIS COURSE

8. Time Intelligence and Date Dimensions

- Creating and managing a date dimension to:
  - Analyze trends over time
  - Implement time-based calculations
  - Use functions such as USERELATIONSHIP for multiple date relationships

WHAT YOU WILL LEARN IN THIS COURSE